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Active Management Strategies for Localized Prostate Cancer

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Keywords

localized prostate cancer; risk stratification; step-wise decision

Questions

1. Decisions regarding the benefit of prostate cancer treatment should be primarily rooted in:
    a. life expectancy calculation of greater than 10 years.
    b. the expertise of the managing physician.
    c. assessment of the risk of prostate cancer mortality relative to other competing risks for death over time.
    d. the prostate-specific antigen (PSA) level at time of diagnosis.
    e. selection of candidates with low risk for metastatic progression.
2. Appropriate candidate selection for prostate cancer treatment may be improved through:
    a. use of validated comorbidity indices and life tables.
    b. exclusion of men who are obese.
    c. exclusion of men over the age of 70.
    d. inclusion of men with high-risk disease features.
    e. psychosocial assessments with validated instruments.

3. The instrument that considers comorbidities with prostate cancer patients in making predictions of 10-year longevity is the:
    a. Kattan nomogram.
    b. PCPT calculator.
    c. WHO Life Tables.
    d. CAPRA score.
    e. prostate cancer-specific Charlson Comorbidity Index (CCI).
4. Men with a high risk of metastatic progression or mortality within 10 years of diagnosis:
    a. are not likely to be helped by active treatment.
    b. are unlikely to develop symptoms of prostate cancer progression.
    c. may benefit from curative-intent treatment, regardless of age.
    d. are most often more likely to die of competing risks.
    e. should empirically be offered androgen deprivation.
5. Which of the following increases the risk of early urinary toxicity after external beam radiation therapy to the prostate?
    a. Previous urinary tract infection
    b. Nocturia
    c. History of hematospermia
    d. Previous transurethral resection of prostate (TURP)
    e. Prior urethroplasty
6. Decision regret in prostate cancer therapy selection is reduced by:
    a. robotic surgery.
    b. less aggressive therapy.
    c. second-opinion consultation.
    d. shared decision-making.
    e. physician bias.
7. The 2005 ISUP modifications in prostate cancer grading resulted in:
    a. characterization of cribriform pattern as Gleason grade 4.
    b. downward migration in Gleason score.
    c. characterization of intraductal variant as Gleason grade 3.
    d. limits on number of cancers deemed Gleason grade 5.
    e. creation of Gleason grade groups.
8. In its initial description, the UCSF-CAPRA score was able to widely discriminate:

a. risk of extraprostatic extension.
b. likelihood of Gleason upgrade.
c. risk of mortality from competing causes.
d. recurrence-free survival.
e. response to adjuvant therapy.

9. The Scandinavian Prostate Cancer Group Study Number 4, comparing radical prostatectomy and watchful waiting, demonstrated improved survival among men undergoing radical prostatectomy. The study was limited by:
    a. predominant inclusion of indolent, low-risk disease.
    b. high rates of death from competing causes.
    c. increased age at time of accrual.
    d. excessive complications from surgery.
    e. primary enrollment in the pre-PSA era.
10. The PIVOT trial, comparing radical prostatectomy and watchful waiting, demonstrated no difference in survival among men undergoing radical prostatectomy or watchful waiting. The study was limited by:
    a. predominant inclusion of indolent, low-risk disease.
    b. high rates of death from competing causes.
    c. increased age at time of accrual.
    d. excessive complications from surgery.
    e. primary enrollment in the pre-PSA era.
11. The Prostate Testing for Cancer and Treatment (ProtecT) trial, a randomized trial of prostatectomy, external-beam radiation with androgen deprivation, or prostate-specific antigen–based active monitoring, demonstrated no difference in mortality between the three study groups at 10 years of follow-up. The study was limited by:
    a. predominant inclusion of indolent, low-risk disease.
    b. high rates of death from competing causes.
    c. increased age at time of accrual.
    d. excessive complications from surgery.
    e. primary enrollment in the pre-PSA era.
12. Whole-gland prostate ablation with high-intensity focused ultrasound is characterized by:

a. demonstration of durable disease control at 15 to 20 years of follow-up.
b. rates of erectile dysfunction comparable to radical prostatectomy.
c. substantial requirement for re-treatment within 5 years.
d. higher rates of erectile dysfunction than cryoablation.
e. urinary toxicity comparable to radiation therapy.

13. In risk assessment, the prostate MRI parameter most predictive of disease outcomes is:
    a. T2-weighted signal attenuation.
    b. diffusion weighted image.
    c. contrast wash-in.
    d. apparent diffusion coefficient (ADC).
    e. hemorrhage on T1-weighted image.
14. Initial risk stratification and staging workup for a man with newly diagnosed clinically localized intermediate risk prostate cancer (T2b, GG 2, PSA 8) includes:
    a. imaging is not indicated in this patient.
    b. cross-sectional abdominopelvic imaging.
    c. bone scan and cross-sectional abdominopelvic imaging.
    d. bone scan and cross-sectional abdominopelvic imaging only if a nomogram predicts greater than 10% probability of pelvic lymph node involvement.
    e. bone scan, cross-sectional abdominopelvic imaging, and fluciclovine PET.
15. Prediction of seminal vesical invasion by prostate MRI is characterized by a high:
    a. false-positive rate.
    b. positive predictive value.
    c. sensitivity.
    d. negative predictive value.
    e. diagnostic accuracy.

Answers

1. **c. Assessment of the risk of prostate cancer mortality relative to other competing risks for death over time.** Evaluation of the relative benefit of prostate cancer treatment in the individual patient should be based upon a determination of the relative probability of prostate cancer metastatic progression and cancer-specific mortality, as compared to the risk of death from other causes, within the expected natural longevity of the patient. A number of tools are available for assessing the risk of prostate cancer progression including nomograms, predictive models, and long-term observational cohorts. Most of these are based upon biopsy characteristics (including Gleason score and disease volume), serum PSA, imaging characteristics, genomic markers, and metastatic staging, in combination, rather than individually. The optimal candidate likely to benefit from treatment is the individual who has a reasonable likelihood of metastatic progression within his lifetime, if not treated. A number of tools exist for predicting life expectancy on the basis of age, adjusted for baseline comorbidities. While a life expectancy of 10 years is often used as a benchmark for benefit from therapy, life expectancy should be considered within the context of disease aggressiveness and the likelihood of disease progression within 5, 10, 15 years and beyond.
2. **a. Use of validated comorbidity indices and life tables.** The perception of longevity in clinical practice is often based solely upon age rather than comorbidity. Daskivich et al. have previously demonstrated that among men with low-risk prostate cancer diagnosed in the VAMC, treatment rates were higher for men with CCI ≥ 3 than those with age greater than 75 (54% vs. 16%), despite a much higher likelihood of other cause mortality within 10 years (70% vs. 24%). Additionally, among men in the SEER-Medicare database, aggressive treatment for men with less than 10 years life expectancy, based upon CCI, was dependent upon age, with 68% to 69% of men aged less than 75 receiving aggressive therapy compared to only 57% and 24% of those aged 75 to 79, or greater than or equal to 80. Similarly, Bratt et al. demonstrated that among men with high-risk prostate cancer in the National Prostate Cancer

Register of Sweden, 50% of men aged less than 70 received radical treatment, as compared to only 10% of men aged 70 to 80, despite a 52% likelihood of 10-year longevity. Improvements in appropriate candidate selection for treatment, and secondarily survival benefit, can be achieved through age-adjusted assessment of longevity on the basis of validated comorbidity indices and life tables.

3. **e. Prostate cancer-specific Charlson Comorbidity Index (CCI).** The most widely utilized scale of comorbidity is the CCI. The CCI was designed to weigh comorbidities in prediction of one-year mortality among inpatients, but ultimately has been shown to be a significant prognostic indicator in the treatment of many types of cancer. In 2011, Daskivich et al. created a prostate cancer-specific CCI (PCCI) weighting comorbidities to predict 10-year risk of mortality among prostate cancer patients. When adjusting for age, and applying the PCCI to men with prostate cancer in the Veteran's Administration health care system (VAMC), segregated PCCI score at 0, 1–2, 3–4, 5–6, 7–9, and 10, the 10-year cumulative incidence of other cause mortality was 10%, 19%, 35%, 60%, 79%, and 99%, respectively.
4. **c. May benefit from curative-intent treatment, regardless of age.** Longevity assessment must be contemplated within the context of cancer aggressiveness. Men with aggressive malignancies may benefit from diagnosis and treatment regardless of a shortened longevity, particularly if presenting at an advanced stage. In men with high-risk, poorly differentiated prostate cancer, within the SEER-Medicare dataset, when comparing cancer-specific survival, aggressive treatment resulted in mortality improvements of 11.3%, 7.9%, 8.6%, and 2.8% at 10-year follow-up for men with Charlson scores of 0, 1, 2, and 3+, respectively, and numbers needed to treat to prevent 1 cancer death at 10 years were 9, 13, 12, and 36 men. Lead time from diagnosis to symptoms and eventual cancer-related mortality is heavily influenced by Gleason score and tumor stage, and this suggests that the relevance of expected longevity in treatment decision-making must be framed within a temporal estimate of cancer progression and mortality risk. Men with a high risk of metastatic progression or mortality within 10 years of

diagnosis should be considered for aggressive therapy despite the presence of one or more competing risks for mortality within the same time period.

5. **d. Previous transurethral resection of prostate (TURP).** For men who have had a previous aggressive TURP for outlet obstruction, the risk of *urinary toxicity* may be higher in those that undergo radiotherapy, but the rates of long-term severe urinary symptoms may be lower. Delay of radiation for 6 to 8 weeks after TURP may reduce the risk of incontinence and bladder neck contracture.
6. **d. Shared decision-making.** Shared decision-making involves a collaborative process in which patients and clinicians make decisions together. Given the controversy surrounding the relative benefit of prostate cancer screening and treatment, shared decision-making, both with regard to PSA testing and prostate cancer treatment, has been strongly advocated by a number of governing bodies. This shared decision-making model has been adopted into guidelines on localized prostate cancer management supported by the AUA, ASTRO, and the SUO. More recently, consensus has been achieved regarding strategies for implementation of shared decision-making in clinical practice. In a recent multicenter, prospective, observational study of 454 men undergoing prostate cancer counseling and treatment, regardless of whether men expressed an initial desire to be actively or passively involved in decision-making or not, after treatment, men who perceived their involvement in decision-making as active were more well-informed with regard to the disease and expressed less decisional regret in follow-up after treatment. The available evidence suggests that interventions aimed at informing patients regarding the comparative effectiveness of prostate cancer management alternatives should include the influence of physician recommendations and family members' desires on patient decision-making. The urologist's recommendation for treatment, however, continues to have the most influence on the decision-making.
7. **a. Characterization of cribriform pattern as Gleason grade 4.** In 2005, the International Society of Urologic Pathology (ISUP) recommended against the use of patterns 1 and 2, leaving a

shortened scoring scale of 6 to 10. Additionally, the consensus recommended that a number of variant patterns previously graded as 3, and all cribriform patterns, should be considered pattern 4. In doing so, there has been an observation of Gleason scoring migration, with a greater number of cancers designated as Gleason score 3+4 or 4+3, as compared to historical cohorts. This migration has led to a reassessment of the accuracy of Gleason scoring in contemporary practice, with most such evaluations demonstrating improved prognostic accuracy. While the grading system appears to be more prognostically accurate, it has created difficulty in comparing historical outcomes of treatment to contemporary cohorts, given the difference in what is meant by Gleason 3+3 and 3+4/4+3.

8. **d. Recurrence-free survival.** The UCSF-Cancer of the Prostate Risk Assessment (CAPRA) score was developed for predicting clinical outcomes using age, PSA level, clinical stage, biopsy Gleason score, and percentage of positive biopsy cores. In its initial description, the UCSF-CAPRA score was able to widely segregate risk from a 5-year recurrence free survival of 85% among low-risk to 8% among high-risk patients. This stratification scheme has subsequently been shown to additionally predict prostate-cancer-specific mortality independently of the treatment given.
9. **e. Primary enrollment in the pre-PSA era.** SPCG-4 was the first to publish a randomized intention-to-treat analysis comparing watchful waiting and radical prostatectomy using informative clinical endpoints such as overall and cancer-specific survival. Furthermore, the SPCG-4 offered the long-term follow-up necessary to characterize between-group differences in survival and employed rigorous study design methods such as blinded histopathologic review. However, because this trial enrolled men predominantly during the pre-PSA era, nearly 90% of the 695 participants harbored palpable disease. Although a number of important observations regarding the comparative effectiveness of radical prostatectomy in the treatment of localized prostate can be made from the SPCG-4 data, this trial does not reflect current prostate cancer screening, detection, and treatment paradigms.

10. **b. High rates of death from competing causes.** The follow-up of the original PIVOT analysis was published in 2017, after 19.5 years of follow-up, demonstrating an absolute risk reduction (ARR) of 5.5% (95% confidence interval [CI]; −1.5% to 12.4%, $P = .06$) for all-cause mortality (61.3% vs. 66.8%) through prostatectomy. An ARR of 4.0% (95% CI; −0.2 to 8.3%, $P = .06$) was found for prostate cancer-specific mortality (7.4% vs. 11.4%). On subgroup analysis, surgery reduced all-cause mortality among men with intermediate-risk disease (ARR = 14.5%, 95% CI 2.8% to 25.6%). By contrast, no benefit was seen in the low- or high-risk groups. While surgery reduced the need for treatment of progressive disease, there were increased incidences of therapy-related long-term complications such as urinary incontinence and erectile and sexual dysfunction. One of the major concerns regarding the PIVOT study design was that the inclusion criterion for the study was life expectancy of greater than 10 year. However, according to the analysis of all-cause mortality after 10-year follow-up, nearly 40% of the study population had died, with only approximately 5% from prostate cancer. After 15 years, up to 60% had died, with ~10% from prostate cancer. These considerations suggest poor risk assessment at baseline inclusion and lessen the impact of the study. Since the design of the study, it is now well understood that patients with low-risk prostate cancer and comorbidities do not benefit within 10 years from radical interventions.
11. **a. Predominant inclusion of indolent, low-risk disease.** In the ProtecT trial, a total of 82,429 men 50 to 69 years of age received a PSA test; 2664 received a diagnosis of localized prostate cancer, and 1643 agreed to undergo randomization to active monitoring (545 men), surgery (553 men), or radiotherapy (plus 3 to 6 months androgen deprivation therapy, 545 men). The primary outcome was prostate-cancer mortality at a median of 10 years of follow-up. In total, 62% of eligible men accepted randomization, comprising a cohort with mostly low- to intermediate-risk tumors. The primary outcome of prostate cancer mortality was not different between the three arms at a median follow-up of 10 years. However, the incidence of metastases was higher in the active monitoring group.

The majority of included patients had a low PSA and Gleason 6 tumors (grade group 1). As the chance of low-risk patients dying from prostate cancer within 10 years is very low, longer follow-up may be needed to draw firm conclusions. The mortality in the study was approximately 1% which is low in comparison to the predicted mortality of 15% at the time the study was designed, thereby reducing the power of the study to assess its intended purpose.

12. **c. Substantial requirement for re-treatment within 5 years.** The primary attraction of whole-gland ablation is the potential to preserve sexual and urinary function more frequently than radical prostatectomy or radiation therapy. While whole-gland cryoablation almost always results in erectile dysfunction, preservation of function with whole-gland high-intensity focused ultrasound (HIFU) is more common, but may vary with the aggressiveness and oncologic efficacy of the procedure. Additionally, the need for re-treatment is 29% to 38% with an increased rate of erectile dysfunction and stricture following a second treatment. As such, men considering whole-gland ablation by HIFU should be counseled with regard to the paucity of long-term outcomes and the potential need for re-treatment.
13. **d. Apparent diffusion coefficient (ADC).** The apparent diffusion coefficient, measured on diffusion-weighted imaging, has been shown to correlate with clinical outcomes and thus can be used as a risk stratification tool. Recent data have explored the value of MRI in monitoring tumors in patients on an active surveillance protocol. In patients with low-risk, localized disease, tumor ADC on diffusion-weighted imaging (DWI) may be a useful marker of prostate cancer progression and may help to identify patients who stand to benefit from radical treatment. As a preoperative imaging tool, tumor ADC and PI-RADS may also be useful to predict biochemical recurrence after radical prostatectomy and radiation therapy. MRI data improves the prediction of biochemical failure within nomograms after external-beam radiation therapy. Pretreatment MRI findings have also been found to predict for biochemical recurrence in intermediate- and high-risk prostate

cancer patients treated with combination brachytherapy and external-beam radiotherapy.

14. **d. Bone scan and cross-sectional abdominopelvic imaging only if a nomogram predicts greater than 10% probability of pelvic lymph node involvement.** Guidelines for initial diagnostic imaging in the newly diagnosed prostate cancer patient vary considerably. While most definitions support radionuclide bone scan for the initial staging of high-risk prostate cancer, not all specifically address imaging for soft tissue metastases. Those that do commonly endorse pelvic CT or MRI. Current NCCN guidelines recommend the use of technetium 99m bone scan and cross-sectional imaging (CT or MRI) among high-risk and unfavorable intermediate-risk men deemed to be at high risk for occult metastasis on the basis of clinical parameters or nomograms. In cases of equivocal bone scans, PET/CT or MRI have been endorsed by several professional organizations.
15. **b. Positive predictive value.** For identifying extraprostatic extension (EPE) or seminal vesical invasion (SVI), MRI is the only examination with sufficient accuracy to be used in clinical practice given its combination of both high spatial and contrast resolution. Historically, assessment for EPE on MRI was based on T2-weighted imaging (T2WI) which provides the best spatial resolution of the MR sequences typically employed. The identification of EPE on T2WI alone is dependent on the amount of EPE present. Gross extension or disruption of the capsule is highly predictive of EPE, but more subtle findings such as irregularity or bulging of the “capsule” have a lower predictive value. As such, when deemed positive for seminal vesical invasion, MRI is generally correct resulting in low rates of false-positive and high positive predictive value. In general, MRI is poor for the detection of focal ECE, as demonstrated by the majority of patients with ECE, resulting in poor sensitivity and low negative predictive value. More subtle findings, such as linear length of capsular contact at the region of suspicion, may improve the pre-treatment prediction of the likelihood of ECE. Accuracy for more subtle EPE is improved when T2WI is combined with diffusion-weighted

imaging (DWI) or dynamic contrast enhanced (DCE) imaging primarily because these additional sequences improve dominant tumor localization, which in turn helps predict presence of EPE, however, uncertainty remains because these ancillary sequences have lower spatial resolution than T2WI.

Chapter Review

1. Prostate cancer treatment outcomes may be influenced by the malignant potential of the tumors as well as by the treatment used. Furthermore, outcome measurements are not necessarily comparable among different forms of therapy, confounding comparisons.
2. Traditionally, deferred treatment has been reserved for men with a life expectancy of less than 10 years and a low-grade prostate cancer. Additional research is needed to define the parameters for safe use of active surveillance in younger men, including the appropriate selection criteria, follow-up procedures, and trigger points for intervention.
3. A prospective, randomized clinical trial reported that patients with clinically localized prostate cancer managed with watchful waiting have significantly higher rates of local cancer progression, metastases, and death from prostate cancer than do those treated initially with radical prostatectomy. In addition, as discussed earlier, a prospective, randomized screening trial reported that prostate cancer screening with PSA and DRE reduces prostate cancer–specific mortality by 20% to 27% with early follow-up and up to 44% with longer follow-up. Because many of these patients were treated with radical prostatectomy, it may also be inferred to be evidence of the efficacy of radical prostatectomy for localized prostate cancer.
4. Radical prostatectomy was the first treatment used for prostate cancer, and it still remains the gold standard. An ideal candidate for radical prostatectomy is a healthy man with a life expectancy of at least 10 years. Preoperative clinical and pathologic parameters are

often used to predict the pathologic stage and thus to identify patients most likely to benefit from the operation.

5. A rising serum PSA level is usually the earliest evidence of tumor recurrence after radical prostatectomy and is frequently an intermediate end point for treatment outcomes. However, not all patients with biochemical recurrence ultimately develop metastases or die of prostate cancer.
6. The most common late complications of radical prostatectomy are erectile dysfunction, urinary incontinence, hernia, and urethral stricture. The return of erectile function after surgery correlates with age of the patient, preoperative potency status, extent of nerve-sparing surgery, and era of surgery; the return of urinary continence is associated with the patient's age.
7. The long-term outcome of cancer control is better documented for open prostatectomy.
8. External beam radiotherapy uses gamma radiation beams directed at the prostate and surrounding tissues through multiple fields. To minimize radiation injury to the bladder and the rectum, 3D-CRT, IMRT, image-guided radiotherapy, SBRT, and proton beam radiotherapy have been developed. Patients with a high PSA level, high Gleason score, or large-volume tumor benefit from ADT in conjunction with radiotherapy. Currently, the most frequently used definition to determine treatment success after radiation therapy is the Phoenix definition, which requires PSA increases of 2 ng/mL above the nadir PSA level.
9. With brachytherapy, radioactive seeds or needles are implanted directly into the prostate gland to deliver a high dose of radiation to the tumor while attempting to spare the bladder and the rectum. Brachytherapy is used primarily for the treatment of patients with clinically localized prostate cancer, but it is seldom used for the treatment of high-volume, high-risk prostate cancers. Urinary symptoms are more common after brachytherapy than after external beam radiotherapy, especially in patients with prostatic hyperplasia.
10. Adjuvant radiotherapy shortly after surgery is most likely to benefit patients with extensive positive surgical margins or extracapsular tumor extension without seminal vesicle invasion or lymph node

involvement. Patients most likely to have favorable responses to salvage radiotherapy are those with PSA recurrence long after surgery, slowly rising PSA level, low-grade tumor, and no seminal vesicle invasion or lymph node metastases. However, mortality benefits have also been demonstrated in patients with rapid PSA doubling times and high-risk tumors. Adjuvant radiotherapy reduces relapse rates in patients with high-risk tumor features but does not increase overall survival. Early salvage radiotherapy produces durable responses in most patients with intermediate-risk tumor features. It is unknown whether early adjuvant radiation is better than delayed salvage therapy in patients with adverse pathology findings after radical prostatectomy.

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Active Surveillance of Prostate Cancer

Laurence Klotz

Keywords

active surveillance; biomarkers; expectant management; imaging; prevention; prostate cancer

Questions

1. Which of the following is true about low-risk prostate cancer?
    a. The major concern is biological grade progression over time (from GG 1 to GG 2-5).
    b. About 25% of patients harbor occult higher-grade cancer at diagnosis.
    c. 30% of middle-aged Caucasian men harbor some prostate cancer.
    d. Most low-risk prostate cancers harbor significant genetic mutations.
    e. GG 1 cancer metastasizes in 3% to 5% of cases.
2. In men with low-risk prostate cancer on active surveillance:
    a. With long-term follow-up, local progression of disease resulting in outflow obstruction, retention, and hydronephrosis is common.
    b. Delayed treatment is less likely to be effective.

c. Approximately 40% will have a signal for treatment by 10 years.
d. The likelihood of metastasis of low-grade prostate cancer is 0.5% to 5% at 15 years.
e. A PSA above 10 is a significant risk factor for metastatic disease.

3. Clinical parameters which predict for the likelihood of upgrading include:
    a. Extent of core involvement.
    b. Bilaterality.
    c. PSA density.
    d. Proportion of cores involved.
    e. All of the above.
4. Which of the following parameters should preclude active surveillance?
    a. Extensive GG1 disease.
    b. High PSA density.
    c. Baseline PSA >10.
    d. Presence of any Gleason pattern 4.
    e. None of the above.
5. The switch from active surveillance to watchful waiting should be based on:
    a. More than 10 years stable on surveillance.
    b. More than 2 successive negative biopsies and negative MRI.
    c. An estimated life expectancy of <5 to 7 years.
    d. Presence of cardiovascular co-morbidity.
    e. Patient refuses further biopsies.
6. For patients on surveillance, the trigger for intervention should be:
    a. Short PSA doubling time (<3 years) or PSA velocity >2.0/year.
    b. Progression in lesion size or conspicuity on MRI.
    c. Grade group progression.
    d. Volume progression of GG 1.
    e. Tissue-based biomarker showing >15% risk of high-grade cancer.

7. Most national guidelines (Cancer Care Ontario, AUA, ASCO) state which of the following regarding active surveillance?
    a. For low-risk disease, it is one of several recommended options.
    b. Patients should be followed with serial PSA, DRE, and biopsy.
    c. MRI is a useful diagnostic test in these patients and should be incorporated into routine clinical practice.
    d. Tissue-based biomarkers have a clear role in the low-risk patient in identifying those at risk.
    e. Patients with intermediate risk disease should not be offered surveillance.

Answers

1. **b. About 25% of patients harbor occult higher-grade cancer at diagnosis.** Grade progression over time occurs in 1% to 2% per year; this is much less of a concern than coexistent high-grade cancer. Fifty to sixty percent of middle-aged men have microfocal prostate cancer. Most low-risk cancers have a very low mutational burden. GG1 cancer is a non-metastasizing lesion; low-grade cancers in men with metastasis invariably have an underlying higher-grade cancer.
2. **c. Approximately 40% will have a signal for treatment by 10 years.** Local progression of cancer is rare. Delayed treatment in this cohort does not reduce effectiveness of therapy. In published series, 20% to 30% are treated by 5 years and 30% to 40% by 10 years. Metastasis occurs from higher-grade cancer (which may be occult). PSA is not a strong predictor for metastasis.
3. **e. All of the above.** These are all clinical parameters that predict for the likelihood of upgrading.
4. **e. None of the above.** None of these parameters should exclude a patient from surveillance. Small amounts of pattern 4 may represent artefactual upgrading and is not an exclusion.
5. **c. An estimated life expectancy of <5 to 7 years.** A patient refusing biopsy may still be followed actively with, for example,

MRI.

6. **c. Grade group progression.** None of the other criteria have been demonstrated to predict for an adverse outcome with reliable accuracy.
7. **b. Patients should be followed with serial PSA, DRE, and biopsy.** It is THE recommended option for low-risk disease; MRI and biomarkers are not recommended routinely; and AS can be offered to selected intermediate-risk patients.

Chapter Review

1. Microfocal Gleason pattern 3 occurs in most men with age, in all races and regions of the world.
2. The mutational burden of Gleason 3 is very favorable, and dramatically different from higher-grade cancer. Most oncogenic pathways characterized by specific genetic alterations are normal in Gleason pattern 3 and aberrant (deleted/amplified/mutated/overexpressed, etc.) in Gleason 4 and 5.
3. Gleason 6 (Grade Group 1) has no metastatic potential.
4. Cancer deaths reported in men with low-risk cancer managed conservatively are thought to be due to the co-existence of higher-grade cancer missed on the diagnostic biopsy.
5. Misattribution of grade (25% to 30% of cases) is much more common than true biological progression from Gleason 3 to 4 or 5 (1% to 2.3% per year).
6. Factors that increase the risk of coexistent higher-grade cancer include PSA density (>0.15), race (African American), and volume of cancer on biopsy (number of positive cores, extent of core involvement).
7. Newly diagnosed men on active surveillance should have a confirmatory biopsy within 1 year, targeting the undersampled areas of the prostate.
8. Following this, they should have PSA q 6 months, and periodic repeat biopsy (every 3 to 5 years).

9. MRI and targeted biopsy may replace serial systematic biopsies in very low-risk men. Since the NPV of MRI is a function of underlying risk, those at higher risk still warrant systematic biopsies.
10. Clinicians managing patients on AS should use the opportunity of regular follow-up visits to counsel patients about the benefits of dietary and lifestyle modifications.

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Open Radical Prostatectomy

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Keywords

localized prostate cancer; radical prostatectomy; radical retropubic prostatectomy

Questions

1. What is the arterial blood supply to the prostate?
    a. The pudendal artery.
    b. The superior vesical artery.
    c. The inferior vesical artery.
    d. The external iliac artery.
    e. The obturator artery.
2. What vessels are located in the neurovascular bundle?
    a. Capsular arteries and veins.
    b. Pudendal artery and vein.
    c. Hemorrhoidal artery and vein.
    d. Santorini plexus.
    e. Accessory pudendal artery.
3. A radical prostatectomy may compromise the arterial blood supply to the penis by injuring the aberrant blood supply from which artery?
    a. The obturator artery.
    b. The inferior vesical artery.

c. The superior vesical artery.
d. The penile artery.
e. All of the above.

4. The main parasympathetic efferent innervation to the pelvic plexus arises from:
    a. S1.
    b. S2 to S4.
    c. T11 to L2.
    d. L3 to S1.
    e. T5 to T8.
5. What is the relationship of the neurovascular bundle to the prostatic fascia?
    a. Inside Denonvilliers fascia.
    b. Outside the lateral pelvic fascia.
    c. Inside the prostatic fascia.
    d. Between the layers of the prostatic fascia and the levator fascia.
    e. Both inside and outside the prostatic fascia.
6. Which anatomic structure is responsible for the maintenance of passive urinary control after radical prostatectomy?
    a. Bladder neck.
    b. Levator ani musculature.
    c. Preprostatic sphincter.
    d. Striated urethral sphincter.
    e. Bulbar urethra.
7. What is the major nerve supply to the striated sphincter and levator ani?
    a. The neurovascular bundle.
    b. The sympathetic fibers from T11 to L2.
    c. The pudendal nerve.
    d. The obturator nerve.
    e. The accessory pudendal nerve.
8. What is the posterior extent of the pelvic lymph node dissection?
    a. The hypogastric vein.
    b. The obturator nerve.
    c. The obturator vessels.

d. The sacral foramen.
e. The pelvic sidewall musculature.

9. In opening the endopelvic fascia, there are often small branches traveling from the prostate to the pelvic sidewall. These branches are tributaries from the:
    a. Obturator artery.
    b. External iliac artery.
    c. Inferior vesical artery.
    d. Pudendal artery and veins.
    e. Neurovascular bundle.
10. How extensively should the puboprostatic ligaments be divided?
    a. Superficially, with just enough incised to expose the junction between the anterior apex of the prostate and the dorsal vein complex.
    b. Extensively, down to the pelvic floor, including the pubourethral component.
    c. Not at all; the puboprostatic ligaments should be left intact.
    d. Widely enough to permit a right angle to be placed around the dorsal vein complex.
    e. Not at all; the puboprostatic ligaments do not need to be divided to perform a radical prostatectomy.
11. When the dorsal vein complex is divided anteriorly, what is the most common major structure that can be damaged, and what is the most common adverse outcome?
    a. Aberrant pudendal arteries; impotence.
    b. Neurovascular bundle; impotence.
    c. Striated urethral sphincter; incontinence.
    d. Levator ani musculature; incontinence.
    e. Both a and b.
12. What is the most common site for a positive surgical margin and when does this occur?
    a. Posterolateral; during release of the neurovascular bundle.
    b. Posterior; when the prostate is dissected from the rectum.
    c. Apex; during division of the striated urethral sphincter-dorsal vein complex.

d. Bladder neck; during separation of the prostate from the bladder.
e. Seminal vesicles.

13. How should the back-bleeders from the dorsal vein complex on the anterior surface of the prostate be oversewn and why?
    a. The edges should be pulled together in the midline to avoid bleeding.
    b. Bunching sutures should be used to avoid excising too much striated sphincter.
    c. The edges should be oversewn in the shape of a V to avoid advancing the neurovascular bundles too far anteriorly on the prostate.
    d. They should be oversewn horizontally to avoid a positive surgical margin.
    e. Oversewing the proximal dorsal vein complex is not required.
14. After the dorsal vein complex has been ligated and the urethra has been divided, what posterior structure, other than the neurovascular bundles, attaches the prostate to the pelvic floor?
    a. Rectourethralis.
    b. Denonvilliers fascia.
    c. Rectal fascia.
    d. Posterior portion of the striated sphincter complex.
    e. Neurovascular bundles.
15. What are the advantages of releasing the levator fascia higher at the apex (more than one answer may be correct)?
    a. More soft tissue on the prostate.
    b. Less traction on the neurovascular bundles as they are released.
    c. Preservation of anterior nerve fibers.
    d. Less blood loss.
    e. Better visualization of the location of the cancer.
16. Once the apex of the prostate has been released, what is the best way to retract the prostate for exposure of the neurovascular bundle?
    a. Traction on the catheter, producing upward rotation of the apex of the prostate.

b. Use of a sponge stick to roll the prostate on its side.
c. Downward displacement of the prostate with a sponge stick.
d. Use of finger dissection to release the prostate posteriorly.
e. Dissection with the sucker.

17. To avoid a positive surgical margin, what is the best way to release the neurovascular bundle?
    a. Right-angle dissection beginning on the posterior surface of the prostate and dissecting anterolaterally.
    b. With sharp dissection, laterally dissecting toward the rectum.
    c. With finger dissection to fracture the neurovascular bundle from the prostate.
    d. With electrocautery to separate the neurovascular bundle from the prostate.
    e. Elevation of the prostate with traction on the Foley catheter.
18. What is the latest point at which a decision can be made regarding preservation or excision of the neurovascular bundle?
    a. When perineural invasion is identified on the needle biopsy specimen.
    b. When the neurovascular bundle is being released from the prostate and fixation is identified.
    c. When the prostate has been removed and tissue covering the posterolateral surface of the prostate is thought to be inadequate.
    d. When the patient is found to have a positive biopsy result at the apex.
    e. When the Partin tables indicate a greater than 50% chance of extraprostatic extension.
19. Before the lateral pedicles are divided, what is the last major branch of the neurovascular bundle that must be identified and released?
    a. Apical branch.
    b. Posterior branch.
    c. Capsular branch.
    d. Bladder neck branch.
    e. Seminal branch.
20. When the vesicourethral anastomosis sutures are being tied, if tension is found, what is the best way to release it?

a. Creating an anterior bladder neck flap.
b. Placing the Foley catheter on traction postoperatively.
c. Releasing attachments of the bladder to the peritoneum.
d. Using vest sutures.
e. Releasing the urethra from the pelvic floor.

21. If there is excessive bleeding from the dorsal vein complex while it is being divided, what should be done?
    a. Abandon the operation and close the incision.
    b. Ligate the hypogastric arteries.
    c. Inflate a Foley balloon and place traction on it.
    d. Divide the dorsal vein complex completely over the urethra and oversew the end.
    e. Deflate the Foley catheter.
22. If a rectal injury occurs during the operation, the most appropriate next step is:
    a. To create a loop colostomy.
    b. To create an end colostomy.
    c. To create a Hartman pouch.
    d. To ensure interposition of the omentum following repair of the injury.
    e. To repair the rectal injury in two layers.
23. In postoperative patients who require transfusions of blood for hypotension, the best approach is to:
    a. Avoid re-exploration because it might damage the anastomosis.
    b. Perform re-exploration.
    c. Place the Foley catheter on traction.
    d. Administer fresh frozen plasma.
    e. Serially monitor the patient in an intensive care unit setting.
24. What is the best way to ensure good coaptation of the anastomotic mucosal surfaces to avoid a bladder neck contracture?
    a. Hold the catheter on traction while tying the sutures.
    b. Use a sponge stick in the perineum.
    c. Use a Babcock clamp to hold the bladder down.
    d. Use vest sutures.
    e. Evert the bladder mucosa.

25. What is the most common cause of incontinence after radical prostatectomy?
    a. Intrinsic sphincter deficiency.
    b. Detrusor instability.
    c. Failure to reconstruct the bladder neck.
    d. Injury to the neurovascular bundles.
    e. Bladder neck contracture.
26. Preservation of the seminal vesicles during radical prostatectomy has demonstrated:
    a. Improved erectile function in most men.
    b. No increase in biochemical recurrence.
    c. Improved early and late urinary control.
    d. Increased rate of pelvic abscess.
    e. None of the above.
27. Preservation of the bladder neck during radical prostatectomy has demonstrated:
    a. Improved erectile function.
    b. Improved long-term urinary control.
    c. Decreased surgical margins.
    d. Improved anastomotic stricture rate.
    e. None of the above.
28. What percentage of men who had bilateral sural nerve grafting demonstrated full erections capable of penetration?
    a. 9%
    b. 13%
    c. 26%
    d. 38%
    e. 57%
29. Sural nerve grafts are placed:
    a. End to end on the ipsilateral side from the tumor.
    b. Above the bladder neck and below the pubic arch.
    c. In reverse to the natural position (proximal to distal and distal to proximal).
    d. In a circle to enhance nerve growth factor release.
    e. Next to the prostatectomy specimen in the pelvis until it is time for anastomosis.

30. Which complication has changed dramatically with experience with salvage prostatectomy?
    a. Overall urinary incontinence.
    b. Potency.
    c. Blood loss.
    d. Rectal injury.
    e. Stricture rate.
31. Concerning postoperative neurapraxia, which of the following statements is TRUE?
    a. The literature supports that it is almost always transient.
    b. It usually results in a motor deficit that is transient.
    c. Most studies show that a self-limited neurapraxia occurs in approximately 25% of patients.
    d. The same rates of neurapraxia tend to occur in retropubic prostatectomy as well.
    e. This is a major source of morbidity and the reason many surgeons do not use this approach.
32. Which of the following statements concerning rectal injury associated with perineal prostatectomy is FALSE?
    a. If unrecognized, it may result in the occurrence of a rectocutaneous or urethrocutaneous fistula.
    b. Despite the close proximity of the rectum in the initial dissection, the incidence is fairly low.
    c. It can be avoided when an assistant places gentle downward pressure on the Lowsley tractor while the rectourethralis muscle is divided.
    d. If repaired with a two-layer closure, most clinical sequelae are avoided.
    e. After repair with a two-layer closure, the operation can continue without a problem.
33. Which of the following statements concerning postoperative care is TRUE?
    a. The diet is rapidly advanced to a regular diet.
    b. Most patients are discharged from the hospital by postoperative day 2.

c. A rectal suppository is administered on a scheduled basis while in the hospital to minimize Foley catheter discomfort except in cases of intraoperative rectal injury.
d. a and b only.
e. All of the above.

34. Which of the following statements concerning the technique of urethral anastomosis is TRUE?
    a. The presence of the Lowsley retractor assists in identifying the membranous urethral stump for the initial placement of interrupted sutures.
    b. A running suture technique is advocated for a watertight anastomosis.
    c. The visualization of the anastomosis is difficult, one of the few disadvantages of the radical perineal prostatectomy.
    d. The sutures are interrupted in a tennis-racquet fashion.
    e. The indwelling Foley catheter is not passed until after the anterior vesicourethral anastomotic sutures are placed and tied down.

Answers

1. **c. The inferior vesical artery** The prostate receives arterial blood supply from the inferior vesical artery.
2. **a. Capsular arteries and veins. The capsular branches run along the pelvic sidewall in the lateral pelvic fascia posterolateral to the prostate, providing branches that course ventrally and dorsally to supply the outer portion of the prostate.** Histologically, the capsular arteries and veins are surrounded by an extensive network of nerves. These capsular vessels provide the macroscopic landmark that aids in the identification of the microscopic branches of the pelvic plexus that innervate the corpora cavernosa.
3. **e. All of the above.** The major arterial supply to the corpora cavernosa is derived from the internal pudendal artery. **However, pudendal arteries can arise from the obturator, inferior vesical,**

and superior vesical arteries. Because these aberrant branches travel along the lower part of the bladder and anterolateral surface of the prostate, they are divided during radical prostatectomy. This may compromise arterial supply to the penis, especially in older patients with borderline penile blood flow.

4. **b. S2 to S4.** The autonomic innervation of the pelvic organs and external genitalia arises from the pelvic plexus, which is formed by parasympathetic visceral efferent preganglionic fibers that arise from the sacral center (S2 to S4).
5. **d. Between the layers of the prostatic fascia and the levator fascia.** The neurovascular bundles are located in the lateral pelvic fascia between the prostatic and levator fasciae.
6. **d. Striated urethral sphincter.** The striated sphincter contains fatigue-resistant, slow-twitch fibers that are responsible for passive urinary control.
7. **c. The pudendal nerve.** The pudendal nerve provides the major nerve supply to the striated sphincter and levator ani.
8. **a. The hypogastric vein.** The obturator artery and vein are skeletonized but are usually left undisturbed and are not ligated unless excessive bleeding occurs. The dissection then continues down to the pelvic floor, exposing the hypogastric veins.
9. **d. Pudendal artery and veins.** The incision in the endopelvic fascia is carefully extended in an anteromedial direction toward the puboprostatic ligaments. At this point, one often encounters small arterial and venous branches from the pudendal vessels, which perforate the pelvic musculature to supply the prostate. These vessels should be ligated with clips to avoid coagulation injury to the pudendal artery and nerve, which are located just deep to this muscle as they travel along the pubic ramus.
10. **a. Superficially, with just enough incised to expose the junction between the anterior apex of the prostate and the dorsal vein complex.** The dissection should continue down far enough to expose the juncture between the apex of the prostate and the anterior surface of the dorsal vein complex at the point where it will be divided. **The pubourethral component of the complex**

must remain intact to preserve the anterior fixation of the striated urethral sphincter to the pubis.

11. **c. Striated urethral sphincter; incontinence.** The goal is to divide the complex with minimal blood loss while avoiding damage to the striated sphincter.
12. **c. Apex; during division of the striated urethral sphincter-dorsal vein complex.** The exact plane on the anterior surface of the prostate can be visualized, avoiding inadvertent entry into the anterior prostate and ensuring minimal excision of the striated sphincter musculature. This is the most common site for positive surgical margins because it can be difficult to identify the anterior apical surface of the prostate.
13. **c. The edges should be oversewn in the shape of a** V **to avoid advancing the neurovascular bundles too far anteriorly on the prostate.** To avoid back-bleeding from the anterior surface of the prostate, the edges of the proximal dorsal vein complex on the anterior surface of the prostate are sewn in the shape of a V with a running 2-0 absorbable suture. **If one tries to pull these edges together in the midline, the neurovascular bundles can be advanced too far anteriorly on the prostate.**
14. **d. Posterior portion of the striated sphincter complex.** The posterior band of urethra is now divided to expose the posterior portion of the striated urethral sphincter complex. The posterior sphincter complex is composed of skeletal muscle and fibrous tissue.
15. **b. Less traction on the neurovascular bundles as they are released, and c. Preservation of anterior nerve fibers.** The purpose of this technique is to speed up recovery of sexual function by reducing traction on the branches of the nerves to the cavernous bodies and striated sphincter and/or avoiding inadvertent transection of the small branches that travel anteriorly. However, because there is less soft tissue at the apex, the risk of positive margins may be increased.
16. **b. Use of a sponge stick to roll the prostate on its side.** When the surgeon releases the neurovascular bundle, there should be no

upward traction on the prostate. Rather, the prostate should be rolled from side to side.

17. **a. Right-angle dissection beginning on the posterior surface of the prostate and dissecting anterolaterally.** After the plane between the rectum and prostate in the midline has been developed, it is possible to release the neurovascular bundle from the prostate, beginning at the apex and moving toward the base, by using the sponge stick to roll the prostate over on its side. Beginning on the rectal surface, the bundle is released from the prostate by spreading a right angle gently. With use of this plane, Denonvilliers fascia and the prostatic fascia remain on the prostate; only the residual fragments of the levator fascia are released from the prostate laterally.
18. **c. When the prostate has been removed and tissue covering the posterolateral surface of the prostate is thought to be inadequate. Clues that indicate that wide excision of the neurovascular bundle is necessary include inadequate tissue covering the posterolateral surface of the prostate once the prostate has been removed, leading to secondary wide excision of the neurovascular bundle.** This last point is important to understand. The surgeon does not have to make the decision about whether to excise or preserve the neurovascular bundle until the prostate is removed, and, if there is not enough soft tissue covering the prostate, one can excise the neurovascular bundle then.
19. **b. Posterior branch.** The surgeon should look for a prominent arterial branch traveling from the neurovascular bundle over the seminal vesicles to supply the base of the prostate. This posterior vessel should be ligated on each side and divided. By this method, the neurovascular bundles are no longer tethered to the prostate and fall posteriorly.
20. **c. Releasing attachments of the bladder to the peritoneum.** The anterior suture is tied initially. There should be no tension. If there is, the bladder should be released from the peritoneum.
21. **d. Divide the dorsal vein complex completely over the urethra and oversew the end. If there is troublesome bleeding from the dorsal vein complex at any point, the surgeon should**

completely divide the dorsal vein complex over the urethra and oversew the end. This is the single best means to control bleeding from the dorsal vein complex. Any maneuver short of this will only worsen the bleeding. To gain exposure for the prostatectomy, one must put traction on the prostate. If the dorsal vein is not completely divided, traction opens the partially transected veins and usually worsens the bleeding.

22. **d. To ensure interposition of the omentum following repair of the injury.** It is wise to interpose omentum between the rectal closure and the vesicourethral anastomosis to reduce the possibility of a rectourethral fistula.
23. **b. Perform re-exploration. Our findings suggest that patients requiring acute transfusions for hypotension after radical prostatectomy should undergo exploration to evacuate the pelvic hematoma in an effort to decrease the likelihood of bladder neck contracture and incontinence.**
24. **c. Use a Babcock clamp to hold the bladder down.** We have found that the use of a Babcock clamp to approximate the bladder neck and urethra while the anastomotic sutures are tied has virtually eliminated bladder neck contractures in our practice.
25. **a. Intrinsic sphincter deficiency.** After radical prostatectomy, incontinence is usually secondary to intrinsic sphincter deficiency.
26. **e. None of the above.** Sparing of the seminal vesicles has not improved incontinence, potency, or margin status, and there have been no reported cases of pelvic abscess.
27. **e. None of the above. Sparing of the bladder neck has not improved incontinence, potency, margin status, or stricture rates.**
28. **c. 26%.** The percentage of men who had bilateral sural nerve grafting and demonstrated full erections (capable of penetration) was 26%.
29. **c. In reverse to the natural position (proximal to distal and distal to proximal).** Sural nerve grafts are placed in reverse to the natural position (proximal to distal and distal to proximal).
30. **d. Rectal injury.** Only rectal injury rates have dramatically changed.

31. **a. The literature supports that it is almost always transient.** Sensory neurapraxia of the lower extremity is reported to occur in approximately 2% of radical perineal prostatectomy cases. However, one study did report an incidence of 25%. This is reported significantly more often than with retropubic prostatectomy. True motor deficits are rare. Because of the transient nature, this is not a major source of morbidity.
32. **c. It can be avoided when an assistant places gentle downward pressure on the Lowsley tractor while the rectourethralis muscle is divided.** Traction on the Lowsley tractor during division of the rectourethralis muscle tents the rectum upward and increases the likelihood of injury. Traction should not be placed until after the rectourethralis muscle is divided. When unrecognized, a fistula may ensue. Although one report showed an incidence of rectal injury in 11% of cases, most series recognize an incidence of 1% to 5%. When the injury is recognized and repaired at the time of occurrence, the operation can continue without a problem.
33. **d. a and b only.** Postoperatively, the diet is advanced rapidly as tolerated, patients ambulate early, and the overwhelming majority of patients are discharged by the second postoperative day. **However, rectal stimulation or manipulation is prohibited in the postoperative period.**
34. **e. The indwelling Foley catheter is not passed until after the anterior vesicourethral anastomotic sutures are placed and tied down.** During placement of the anterior vesicourethral anastomotic sutures, a red rubber catheter is placed transurethrally and used to identify the membranous urethral stump and provide traction on the urethra to assist in placement of the sutures. The red rubber catheter is then removed, and the indwelling Foley catheter is then placed retrograde into the bladder. Simple interrupted sutures are placed for the anastomosis. A tennis-racquet technique is used for bladder neck reconstruction if necessary.

Chapter Review

1. The dorsal vein has three major branches: a superficial branch in the midline and two lateral branches that span over the lateral aspects of the prostate.
2. The prostate is covered with three distinct fascial layers: Denonvilliers fascia, prostatic fascia, and levator fascia.
3. Denonvilliers fascia is most prominent and dense near the base of the prostate and overlying the seminal vesicles and thins dramatically more caudad at its termination at the striated sphincter.
4. Laterally, the prostatic fascia fuses with the levator fascia.
5. Following radical prostatectomy, 15% to 20% of men develop an inguinal hernia. It is usually an indirect inguinal hernia.
6. The final decision whether to preserve the neurovascular bundles is made at surgery.
7. Findings that would indicate a neurovascular bundle should be resected include palpable induration in the lateral pelvic fascia, a neurovascular bundle that appears fixed to the prostate, and insufficient tissue on the posterolateral aspect of the specimen.
8. Thermal energy should never be used on or near the neurovascular bundles.
9. Bladder neck contractures occur in less than 10% of patients.
10. Factors important for recovery of erectile function include patient age, preoperative potency status, and preservation of the neurovascular bundles.
11. Potency improves with time such that in one study, 42% of patients were potent at 3 months and 73% at a year.
12. In a randomized study using sural nerve grafting to preserve potency in patients in whom a neurovascular bundle needed to be sacrificed, there was no difference in those grafted versus those who were not.
13. Salvage radical prostatectomy should only be considered in men who have unequivocally clinically localized prostate cancer.
14. Complications following salvage radical prostatectomy include 50% incontinence, 24% anastomotic stricture, and nearly universal erectile dysfunction, with approximately a 45% recurrence rate at 5 years.
15. Complete excision of the seminal vesicle during radical prostatectomy is recommended for cancer control.

16. Surgery should be deferred for 6 to 8 weeks after biopsy and 3 months after a transurethral resection of the prostate.
17. During pelvic lymphadenectomy, one should preserve the lymphatic tissue covering the external iliac artery, which drains the lower extremity.
18. Accessory pudendal arteries should be preserved.
19. The most common site of a positive margin occurs at the apex, followed by the posterior and the posterolateral prostate.
20. The bladder mucosa should be advanced so that the urethral vesicle anastomosis opposes mucosa to mucosa, is watertight, and is tension free.
21. If postoperative phosphodiesterase type 5 inhibitors are to be used, on demand is preferred to daily dosing.
22. Aberrant pudendal arteries may arise from the obturator, inferior vesical, and superior vesical arteries and should be preserved so as not to compromise the blood supply to the penis.
23. Wide excision of the neurovascular bundle should be considered when there is inadequate tissue covering the posterolateral surface of the prostate once the prostate has been removed. The surgeon does not have to make the decision about whether to excise or preserve the neurovascular bundle until the prostate is removed, and, if there is not enough soft tissue covering the prostate, one can excise the neurovascular bundle then.
24. If there is troublesome bleeding from the dorsal vein complex at any point, the surgeon should completely divide the dorsal vein complex over the urethra and oversew the end.
25. Patients requiring acute transfusions for hypotension after radical prostatectomy should undergo exploration to evacuate the pelvic hematoma in an effort to decrease the likelihood of bladder neck contracture and incontinence.
26. Rectal stimulation or manipulation is prohibited in the postoperative period.

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Laparoscopic and Robotic-Assisted Radical Prostatectomy and Pelvic Lymphadenectomy

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Keywords

laparoscopic prostatectomy; pelvic lymphadenectomy; robotic prostatectomy; salvage prostatectomy

Questions

1. With laparoscopic/robotic prostatectomy, a continuous suture for the vesicourethral anastomosis:
    a. avoids incontinence.
    b. has a high rate of bladder neck contracture.
    c. can be performed with the need for only a single knot.
    d. requires an indwelling catheter for at least 2 weeks.
    e. eliminates the need for a pelvic drain.
2. With laparoscopic/robotic radical prostatectomy, positive margin rates are not influenced by:
    a. surgical technique.
    b. patient selection.
    c. the method of pathologic analysis.
    d. transperitoneal versus extraperitoneal exposure.
    e. tumor grade and stage.

3. Compared with open surgical approaches, laparoscopic/robotic prostatectomy has been consistently shown to decrease:
    a. postoperative pain.
    b. urinary incontinence.
    c. bleeding.
    d. erectile dysfunction.
    e. positive margins.
4. Positive surgical margins with laparoscopic/robotic prostatectomy:
    a. decrease as technical experience is gained.
    b. are rare at the prostatic apex.
    c. occur only when extracapsular disease is present.
    d. are seen most commonly at the prostate base.
    e. can be avoided by using a robotic-assisted approach.
5. An advantage of extraperitoneal versus transperitoneal approach to laparoscopic/robotic prostatectomy is:
    a. faster operating room time.
    b. shorter hospitalization.
    c. increased working space.
    d. avoidance of bowel manipulation.
    e. fewer positive margins.
6. Minimally invasive pelvic lymph node dissection:
    a. is difficult to perform along with robotic radical prostatectomy.
    b. should always be performed transperitoneally.
    c. has an increased risk of thromboembolic complication compared with open approaches.
    d. should only be performed for tumors lower than or equal to Gleason grade 7.
    e. can allow lymph node removal comparable with open surgery.
7. Rectal injury with laparoscopic/robotic radical prostatectomy:
    a. is best avoided by antegrade release of the rectum from the posterior prostate.
    b. is usually from trocar placement.
    c. can be avoided by bluntly dividing Denonvilliers fascia.
    d. should be treated with an immediate diverting colostomy.
    e. is often unrecognized and heals spontaneously.

8. Bleeding during laparoscopic/robotic radical prostatectomy is usually minimal because:
    a. the plane of periprostatic tissue dissection is different than with open surgery.
    b. the dorsal vein complex does not have to be divided.
    c. the pneumoperitoneum tamponades venous bleeding.
    d. suturing is easier than with open surgery.
    e. the Trendelenburg position decreases venous pressure.
9. Robotic assistance with laparoscopy is most useful in:
    a. trocar insertion and removal.
    b. maintaining a steady insufflation pressure.
    c. decreasing operating room costs.
    d. facilitating suturing.
    e. eliminating the need for a tableside assistant.
10. The neurovascular bundle lies within which two periprostatic fascial planes?
    a. Prostate capsule and prostatic fascia.
    b. Prostate capsule and levator fascia.
    c. Prostatic fascia and levator fascia.
    d. Denonvilliers fascia and prostate capsule.
    e. Denonvilliers fascia and endopelvic fascia.
11. Antegrade laparoscopic dissection of the prostate results in less blood loss as compared with the retrograde approach due in part to:
    a. early division of the dorsal venous complex and prostatic pedicles.
    b. early division of the prostatic pedicles and late division of the dorsal venous complex.
    c. less tissue manipulation.
    d. better visualization.
    e. late division of the dorsal venous complex and prostatic pedicles.
12. The higher cost of laparoscopic/robotic-assisted as compared with open radical prostatectomy is mostly a consequence of:
    a. higher blood loss and transfusion rates.
    b. a higher complication rate.
    c. a longer operative time and disposable equipment.

d. longer hospital stays.
e. higher surgical and anesthesia charges.

13. As a consequence of the CO_2 pneumoperitoneum used during minimally invasive prostatectomy, the anesthesia team must be most aware of the potential for:
    a. bleeding and hypotension.
    b. hypoxia and acidosis.
    c. tachycardia and hypertension.
    d. bradycardia and hypotension.
    e. hypercarbia and oliguria.
14. Positive margins at the prostatic apex:
    a. are more common with the robotic-assisted technique as compared with open surgery.
    b. can occur due to protrusion of the posterior prostatic apex beneath the urethra.
    c. can occur more commonly with retrograde versus antegrade dissection of the prostate.
    d. are less common in laparoscopic versus open surgery.
    e. are less common than at the prostatic base.
15. Men who are not candidates for laparoscopic/robotic-assisted laparoscopic radical prostatectomy include those with:
    a. palpable tumors.
    b. history of prior pelvic surgery.
    c. morbid obesity.
    d. uncorrectable bleeding diatheses.
    e. prior neoadjuvant hormonal therapy.
16. The following are true about robotic salvage prostatectomy except:
    a. Higher risk of rectal injury than standard robotic-assisted laparoscopic prostatectomy (RALP).
    b. Higher risk of urinary incontinence than standard RALP.
    c. Need to rule out metastatic disease prior to performing.
    d. Should be attempted early in one's experience.
    e. The Foley catheter should be left in place longer than with a standard RALP.

Answers

1. **c. Can be performed with the need for only a single knot.** The vesicourethral anastomosis may be accomplished using either an interrupted closure or a running continuous suture with a single knot.
2. **d. Transperitoneal versus extraperitoneal exposure.** Comparison of margin status between high-volume centers with the operations performed by experienced surgeons has shown no definitive advantage for one surgical approach over the other in achieving negative surgical margins.
3. **c. Bleeding.** Because most of the blood loss that occurs during radical prostatectomy is from venous sinuses, the tamponade effect from the pneumoperitoneum helps diminish ongoing blood loss during laparoscopic robotic prostatectomy (LRP)/RALP. Blood loss of less than a few hundred milliliters is routinely reported.
4. **a. Decrease as technical experience is gained.** In most series of LRP and RALP, positive margin percentages decrease as greater familiarity with the procedure is obtained.
5. **d. Avoidance of bowel manipulation.** As the extraperitoneal technique avoids violation of the peritoneal envelope, bowel manipulation is avoided. It is for this reason that patients with extensive prior abdominal surgery can undergo successful laparoscopic and robotic prostatectomy by the extraperitoneal route.
6. **e. Can allow lymph node removal comparable with open surgery.** Pelvic lymphadenectomy can be performed by open or laparoscopic techniques with no significant difference in nodal yield.
7. **a. Is best avoided by antegrade release of the rectum from the posterior prostate.** Thorough dissection of the rectum off of the posterior prostate is critical to minimize the risk of rectal injury during subsequent steps such as division of the urethra and dissection of the prostatic apex. With LRP and RALP, sharp and complete incision of the posterior layer of Denonvilliers fascia is

necessary after seminal vesicle dissection to allow adequate mobilization of the rectum.

8. **c. The pneumoperitoneum tamponades venous bleeding.** Because most of the blood loss that occurs during radical prostatectomy is from venous sinuses, the tamponade effect from the pneumoperitoneum helps diminish ongoing blood loss during LRP/RALP. Blood loss of less than a few hundred milliliters is routinely reported.
9. **d. Facilitating suturing.** Most surgeons, however, believe that the robotic technology significantly facilitates suturing (especially for the vesicourethral anastomosis) and other aspects of the surgical dissection.
10. **c. Prostatic fascia and levator fascia.** The neurovascular bundle travels between two distinct fascial planes that surround the prostate, namely, the prostatic fascia and levator fascia.
11. **b. Early division of the prostatic pedicles and late division of the dorsal venous complex.** Because the dorsal venous complex is divided early in the operation and the prostatic pedicles late, there is potentially a greater risk of ongoing bleeding with the retrograde technique. In contrast, during the antegrade neurovascular bundle dissection, the arterial blood supply to the prostate (via the prostatic pedicles) is divided early and the dorsal venous complex is divided near the end of the operation, thus reducing blood loss during the operation.
12. **c. A longer operative time and disposable equipment.** In the study by Link and colleagues (2004), the factors that most influenced overall cost in order of importance included operative time, length of hospital stay, and consumable items (e.g., disposable laparoscopic equipment and trocars).
13. **e. Hypercarbia and oliguria.** The anesthesiologist must be aware of the potential consequences of CO_2 insufflation and pneumoperitoneum including oliguria and hypercarbia.
14. **b. Can occur due to protrusion of the posterior prostatic apex beneath the urethra.** Before division of the posterior urethra, great care must be taken to inspect the contour of the posterior prostatic

apex. In some patients, the posterior prostatic apex can protrude beneath the urethra, resulting in an iatrogenic positive margin if not identified.

15. **d. Uncorrectable bleeding diatheses.** Contraindications to minimally invasive laparoscopic prostatectomy include uncorrectable bleeding diatheses or the inability to undergo general anesthesia due to severe cardiopulmonary compromise.
16. **d. Should be attempted early in one's experience.** These cases are perhaps best performed in high-volume centers by experienced robotic surgeons, as they can be very challenging with a higher risk of intraoperative and postoperative complications.

Chapter Review

1. Regarding patient selection, it is strongly advised that patients with more complex anatomic challenges (e.g., large prostate size, large median lobe, morbid obesity, prior pelvic surgery, after radiation, after transurethral resection of prostate [TURP]) be avoided in a surgeon's early experience with LRP and RALP; however, these patient features are not by themselves absolute contraindications for a minimally invasive approach to prostatectomy.
2. Having a skilled bedside assistant who is well versed in basic laparoscopy, but also specifically in the mechanics, setup, and troubleshooting of the robotic system, can greatly facilitate RALP procedures.
3. Both surgeon and anesthesiologist must be aware of the unique physiologic effects of prolonged pneumoperitoneum with patients in the steep Trendelenburg position, including hypercarbia and acidosis, corneal edema, increased intraocular pressure, and neurapraxia and take proper steps to prevent such complications.
4. Use of a transperitoneal or extraperitoneal approach is based primarily on surgeon preference because there does not appear to be a significant advantage of one approach over the other.
5. The robotic technology facilitates suturing and dissection for surgeons who may not have advanced laparoscopic skills.

6. Identification of the precise margin between the bladder neck and prostate can be accomplished by using a set of physical maneuvers and visual cues.
7. After division of the anterior bladder neck, the presence or absence of a median lobe should be defined before division of the posterior bladder neck.
8. Complete mobilization of the plane between the posterior prostate and rectum is an important step in avoiding rectal injury and defining the medial aspect of the prostatic pedicles and neurovascular bundles (NVBs).
9. Use of thermal energy should be minimized during control of the prostatic pedicles and dissection near the NVBs, as thermal energy has been shown to be deleterious to cavernous nerve function in both animal and human studies.
10. An interfascial dissection of the NVB is preferred in patients with presumed organ-confined cancer who desire NVB preservation for potency.
11. An anterior or posterior protruding lip of prostate should be anticipated when dividing the prostatic apex from the urethra to avoid an iatrogenic positive apical margin.
12. Bladder neck reconstruction may be required if there is a size discrepancy between the bladder neck and urethra, especially in patients with prior TURP, median lobe, or large prostate gland.
13. The vesicourethral anastomosis is most efficiently accomplished with a running continuous suture.
14. Complications and Outcomes of RALP: (1) **Patient positioning**: Care must be taken when positioning in the Trendelenburg position that any potential, area of neural compression is padded. (2) **Blood loss**: Blood loss of robotic prostatectomy is usually minimal, and transfusion requirements should be less than 2%. (3) **Bowel injury**: Injury to the rectum is well reported but rare, and should be repaired when recognized. The rectal injury rate is higher in salvage surgery. (4) **Urine leak**: This occasionally occurs through a non-watertight anastomosis and requires a longer urinary catheter period. (5) **Bladder neck stricture**: The occurrence of bladder neck stricture is much reduced since the introduction of robotic anastomotic

technique. The rate should be less than 5%. (6) **Machine failure**: Machine failure is rare and impacts less than 0.5% of cases. (7) **Oncologic outcomes**: There is equivalence in oncologic outcomes between robotic and open radical prostatectomy. The robotic oncologic outcomes (margin negative) are better when surgery is performed in a high-volume centers. (8) **Potency outcomes**: This is a statistically better outcome for potency recovery following robotic prostatectomy versus the open method. (9) **Continence outcomes**: Even in the best hands, at least 20% of men following robotic prostatectomy will not be pad free, and 5% will have severe incontinence irrespective of technical excellence. (10) **Minimally invasive surgery outcomes.** Following robotic prostatectomy, patients can be discharged in 24 hours and return to normal activity within 3 weeks. This is a significant advantage of the minimally invasive surgical approach.

15. Pelvic lymph node dissection (PLND) is generally recommended in patients with intermediate- or high-risk prostate cancer or by having a >5% risk of nodal metastasis using a clinical nomogram.
16. Although an extended PLND may yield a higher lymph node count, debate continues as to the extent of PLND and the clinical benefit of removing cancerous nodes, given the increased time and complications rates.
17. Use of clips on identifiable lymphatic channels can minimize the occurrence of postoperative lymphoceles.
18. A transperitoneal approach is not protective against the formation of a lymphocele because loculation of lymphatic fluid can still occur within the peritoneal cavity.
19. Symptomatic lymphoceles, which cause local problems such as venous or bladder compression, may require percutaneous or laparoscopic drainage.

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Radiation Therapy for Prostate Cancer

Ryan Phillips, Sarah Hazell,, and Daniel Y. Song

Keywords

brachytherapy; prostate cancer; radiotherapy

Questions

1. When managed with radiation alone, patients with high-risk prostate cancer have a 5-year prostate-specific antigen (PSA) failure-free survival rate of approximately:
    a. 75%.
    b. 60%.
    c. 33%.
    d. 10%.
2. Which of the following represents unfavorable intermediate-risk disease?
    a. Gleason 3 + 4, PSA 9.5, cT2a
    b. Gleason 4 + 3, PSA 9.5, cT3a
    c. Gleason 4 + 4, PSA 9.5, cT1c
    d. Gleason 4 + 3, PSA 12.8, cT2a
3. A 58-year-old man presents with a T2b, Gleason 7 (4 + 3: 80% pattern 4, 2/6 sextants positive, 4/10 cores), PSA 7.8 ng/mL. He is treated with 6 months combined androgen deprivation therapy (ADT) + 78 Gy intensity-modulated radiation therapy (IMRT).

Testosterone recovers 6 months following completion of IMRT. At 12 months, PSA is 0.8 ng/mL; at 18 months, 2.6 ng/mL; at 24 months, 3.2 ng/mL; and now at 30 months, following IMRT, PSA is 5.4 ng/mL. The best next step in management is:

a. because the most likely explanation for the rising PSA is testosterone recovery, reassure the patient and continue to follow up.
b. arrange biopsy for presumed local failure.
c. because patient clearly has failed radiotherapy, advise radical local salvage.
d. because patient's PSA doubling time and time to biochemical failure make him an unlikely candidate for local salvage because they are indicative of systemic failure, arrange systemic staging with or without multiparametric magnetic resonance imaging (MRI) of the prostate.
e. The PSA kinetics are compatible with a benign increase or "bounce." Reassure the patient and continue to monitor.

4. A 53-year-old man presents with T2a, Gleason 7 (3 + 4: 20% pattern 4, 1/6 sextants positive, 2/10 cores), PSA 5.9 ng/mL. He is treated with iodine-125 (^{125}I) permanent seed prostate brachytherapy. At 12 months, PSA has fallen to 1.2 ng/mL. At 15 months, PSA is 1.8 ng/mL; at 18 months, it is 3.2 ng/mL. He is sexually active and asymptomatic. Which is the best next step?

a. Brachytherapy has failed. Start ADT.
b. Brachytherapy has failed. Arrange radical local salvage.
c. Brachytherapy has failed. Arrange biopsy before radical local salvage.
d. Recognizing that this is classic timing for a benign PSA bounce, contact his radiation oncologist, review the dosimetry of the implant, and continue to monitor the PSA every 3 months, planning for eventual biopsy if the PSA has not started to decline again by 30 months.
e. Given his time to nadir of 12 months and PSA doubling time of 4 months, explain to the patient that this is almost certainly distant failure and discuss timing of ADT.

5. In two large trials that randomized patients with high-risk and locally advanced prostate cancer to ADT with versus without radiation therapy, the addition of radiation improved what outcome by an absolute of 8% to 10%?
    a. Biochemical recurrence-free survival
    b. Clinical recurrence-free survival
    c. Metastasis-free survival
    d. Disease-specific survival
    e. Overall survival
6. Modern radiation therapy for prostate cancer differs from the technology used in the 1980s to 1990s in what ways? Modern radiation uses:
    a. computed tomography (CT)-based treatment planning.
    b. intensity-modulated radiation delivery.
    c. image guidance.
    d. dose-escalated radiation.
    e. all of the above.
7. A randomized trial comparing 76 Gy in 38 fractions to 70.2 Gy in 26 fractions found:
    a. the shorter regimen improved 5-year biochemical failure and had comparable rectal and urinary toxicity.
    b. the shorter regimen improved 5-year biochemical failure and had higher late urinary toxicity.
    c. the shorter regimen improved 5-year biochemical failure and had higher late rectal toxicity.
    d. the shorter regimen did not improve 5-year biochemical failure and had higher late urinary toxicity in men with poor baseline urinary function.
    e. the shorter regimen did not improve 5-year biochemical failure and had higher late rectal toxicity.
8. In 2013, radium-223 was approved by the U.S. Food and Drug Administration (FDA) for use in the treatment of men with _____ because of a prolongation in _____.
    a. castrate-resistant metastatic prostate cancer, overall survival
    b. castrate-resistant metastatic prostate cancer, disease-free survival

c. castrate-resistant nonmetastatic prostate cancer, overall survival
d. castrate-resistant metastatic prostate cancer, PSA failure–free survival
e. castrate-resistant nonmetastatic prostate cancer, PSA failure–free survival

Imaging

1. A voiding cystourethrogram on a 72-year-old man who presents with urinary tract infections 2 years after combination external beam radiation therapy and brachytherapy for prostate cancer is depicted in Fig. 157.1. The most likely diagnosis is:

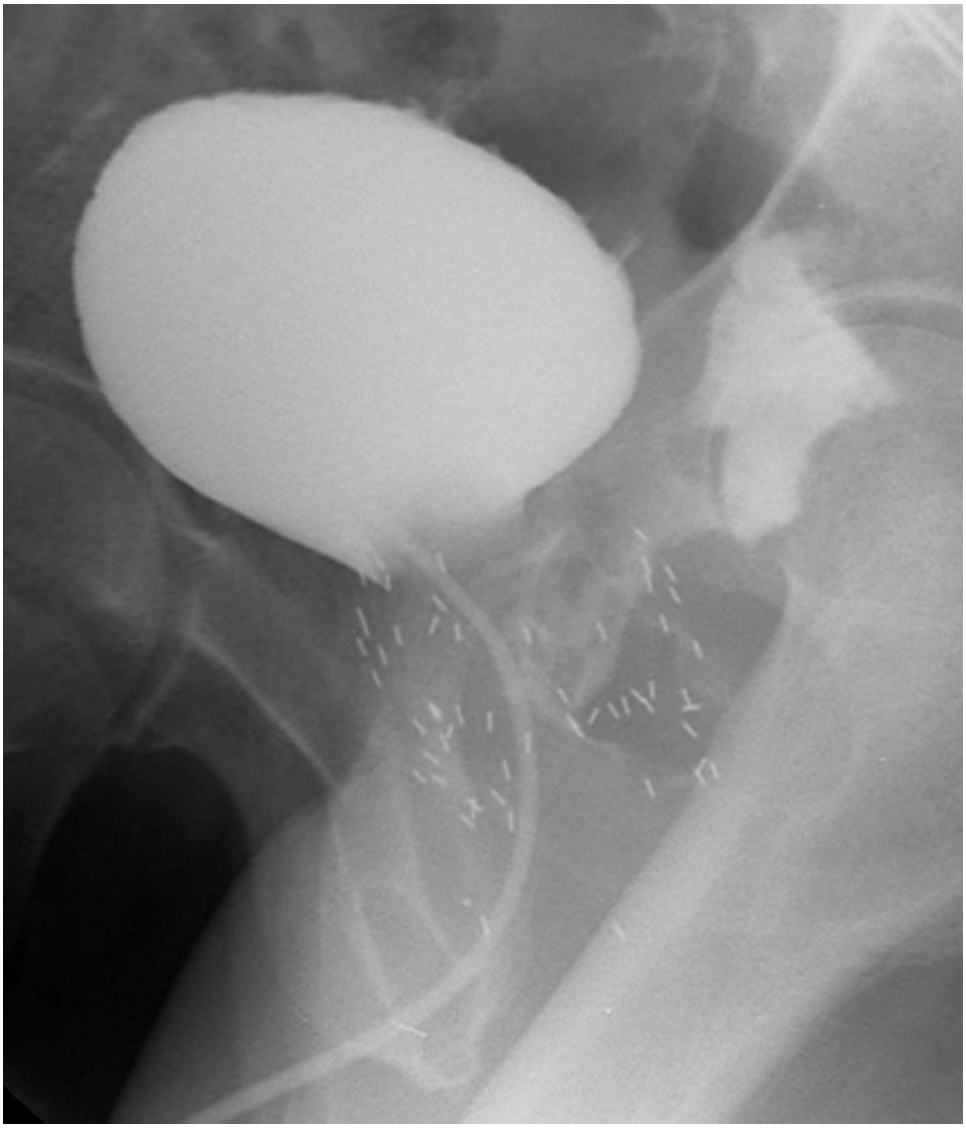


FIG. 157.1

a. brachytherapy seed migration.
b. urethra rectal fistula.
c. vesico sigmoid fistula.
d. urethra cutaneous fistula.
e. vesico rectal fistula.

Answers

1. **c. 33%.** (see D'Amico AV, Whittington R, Malkowicz S, et al: Biochemical outcome after radical prostatectomy, external beam

radiation therapy, or interstitial radiation therapy for clinically localized prostate cancer. *JAMA* 280:969–974, 1998.)

2. **d. Gleason 4 + 3, PSA 12.8, cT2a.** (see Zumsteg ZS, Spratt DE, Pei I, et al: A new risk classification system for therapeutic decision making with intermediate-risk prostate cancer patients undergoing dose-escalated external-beam radiation therapy. *Eur Urol* S0302-2838[13]00257-1, 2013.)
3. **d. Because patient's PSA doubling time and time to biochemical failure make him an unlikely candidate for local salvage because they are indicative of systemic failure, arrange systemic staging with or without multiparametric magnetic resonance imaging (MRI) of the prostate.** This PSA increase is much too high to be due to testosterone recovery. Because this patient was treated with IMRT and not brachytherapy, this is not a benign increase. A 58-year-old man should not be denied potentially curative salvage, but his PSA kinetics are strongly suggestive of distant failure. Although ADT may well be the correct maneuver, systemic staging should be undertaken first. **A posttreatment biopsy of the prostate should be interpretable 30 months after completion of treatment, but even if residual disease is demonstrated, this patient is at very high risk of also harboring systemic disease because of his PSA kinetics and should fully understand this before consenting to further local treatment.**
4. **d. Recognizing that this is classic timing for a benign PSA bounce, contact his radiation oncologist, review the dosimetry of the implant, and continue to monitor the PSA every 3 months, planning for eventual biopsy if the PSA has not started to decline again by 30 months. This is classic timing for a benign PSA bounce after brachytherapy monotherapy**. Bounces are seen in 65% of men younger than 55 years, especially those who are sexually active, and can be quite dramatic, with kinetics characteristic of distant failure because of the early onset and rapid doubling. **Reassurance, close monitoring, biopsy if the bounce has not resolved by 30 months, and systemic staging if the PSA approaches 10 ng/mL are appropriate.**

5. **e. Overall survival.** In the NCIC/MRC trial of 1205 patients, the addition of radiation therapy significantly improved overall survival: 7-year survival 66% for ADT versus 74% ADT/RT (P = .033). In a Scandinavian trial (SPCG-7/SFUO-3) of 875 patients, radiation therapy decreased overall mortality: 10-year mortality was 39.4% for ADT versus 29.6% for ADT/RT ($P < .05$).
6. **e. All of the above.** CT-based treatment planning was developed in the 1990s, IMRT in the early 2000s, and image guidance more recently. Four randomized trials consistently demonstrated improved cancer control with "dose-escalated" radiation, which has changed the standard of care in prostate cancer radiation.
7. **d. The shorter regimen did not improve 5-year biochemical failure and had higher late urinary toxicity in men with poor baseline urinary function.** (See Pollack A, Walker G, Buyyounouski MK, et al: Five year results of a randomized external beam radiotherapy hypofractionation trial for prostate cancer. *Int J Radiat Oncol Biol Phys* 81[2]:S1, 2011.)
8. **a. Castrate-resistant metastatic prostate cancer, overall survival.** (See Parker C, Nilsson S, Heinrich D, et al: ALSYMPCA Investigators. Alpha emitter radium-223 and survival in metastatic prostate cancer. *N Engl J Med.* 369[3]:213–223, 2013.)

Imaging

1. **b. Urethra rectal fistula.** There is partial opacification of the irregular prostatic urethra with immediate filling of the rectum. The bladder is normal in appearance, making option e incorrect. The contrast is in the rectum, making a urethra cutaneous fistula and vesico sigmoid fistula incorrect. The seeds are located in the area one would expect following brachytherapy.

Chapter Review

1. The advantage of brachytherapy is that there is a rapid falloff of dose within a few millimeters of the seed implant.

2. A pretreatment PSA velocity of greater than 2 ng/mL/yr is associated with an increased risk of biochemical failure following radiation therapy.
3. The American Society for Radiation Oncology (ASTRO) definition of failure following radiation therapy is three consecutive rises in PSA following the nadir. It is recommended that these determinations be obtained 3 to 4 months apart. The Phoenix definition of failure is a 2 ng/mL or more increase in PSA above the nadir.
4. A PSA nadir of less than 0.5 ng/mL is correlated with successful treatment.
5. PSA nadir is the strongest predictor of outcome.
6. Approximately 50% of potent men are impotent at 5 years following conformal external beam radiation therapy; 10% will have some rectal bleeding. Less common complications include urinary and fecal incontinence, hemorrhagic cystitis, and urethral stricture.
7. Conformal radiation therapy (CRT) uses a computerized algorithm to conform the dose of radiation to the contours of the prostate.
8. IMRT uses a set of radiation beams with changing intensities distributed across the field.
9. When IMRT is compared with CRT, lower doses are delivered to critical tissues such as rectum, bladder, and small bowel.
10. Heavy particle beams such as neutrons and protons exhibit a Bragg effect. This is manifested by a sharp falloff beyond the particle's tissue range, thus delivering little radiation beyond that point.
11. ^{125}I has a half-life of approximately 60 days; palladium-103 has a half-life of 17 days.
12. Treatment of impending spinal cord compression due to prostate cancer consists of immediate ablation of androgens (most effectively accomplished by bilateral orchiectomy), steroids, and radiation therapy. Occasionally a decompression laminectomy is required emergently.
13. The dose delivered to the prostate with IMRT is 75 to 79 Gy; for brachytherapy with ^{125}I is 140 to 160 Gy; and for ^{103}Pd is 110 to 125 Gy.

14. Pretreatment risk stratification may be divided as follows: low risk—T1c-T2a, PSA less than 11 ng/mL and no Gleason score higher than 6; intermediate risk—T2b, PSA 11 to 20 ng/mL or Gleason score of 7; and high risk—T2c, PSA greater than 20 ng/mL or Gleason score greater than 7.
15. The percentage of positive biopsies and the PSA velocity have also been used for risk stratification.
16. Radiation causes significant changes in cell architecture that can make interpretation of biopsy specimens after radiation therapy difficult.
17. At present there does not appear to be any advantage of proton therapy over conventional photon IMRT.
18. Radium-223 used to treat metastatic prostate cancer to bone is an alpha particle emitter and as such there is less damage to hematopoietic marrow elements when compared with strontium-89.
19. A postradiation treatment biopsy of the prostate should be interpretable 30 months after completion of treatment.

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Focal Therapy for Prostate Cancer

Kae Jack Tay, and Thomas J. Polascik

Keywords

focal therapy; mpMRI; prostate cancer; prostatectomy; radiation; saturation biopsy; targeted biopsy; targeted therapy

Questions

1. Advantages of using mpMRI to identify suitable candidates for prostate focal therapy include all the following EXCEPT:
    a. It improves the detection of anterior zone cancers
    b. It preferentially detects high-grade lesions
    c. It accurately estimates lesion size for determination of ablation margin
    d. It improves the detection of extra-prostatic extension
    e. It helps accurately identify the index lesion
2. Selection of an ablative energy source for prostate focal therapy is predicated on:
    a. availability of ablative technology.
    b. lesion location.
    c. presence of extensive intra-prostatic calcifications.
    d. both (a) and (b).
    e. all of the above.
3. Technical innovations improving focal cryotherapy for prostate cancer include all EXCEPT the use of:

a. transrectal ultrasound.
b. urethral warming catheters.
c. warming of neurovascular bundles.
d. thermocouples placed in critical areas of the prostate.
e. smaller-diameter cryoprobes allowing percutaneous insertion.

4. What is the most clinically important parameter of tissue ablation other than lowest temperature achieved by cryotherapy?
   a. The diameter of the cryoprobe
   b. The number of freeze/thaw cycles
   c. The velocity of tissue thawing
   d. The velocity of tissue freezing
   e. The duration of freezing
5. What is the characteristic appearance of ice ball created during prostate focal cryotherapy on ultrasound?
   a. Mixed echogenicity
   b. Hyperechogenicity
   c. Anechogenicity with a hypochogenic rim
   d. Hypoechogenicity with a hyperechogenic rim
   e. None of the above
6. Prostate cell death is likely to occur completely in a single freeze cycle when tissue temperature reaches:
   a. 20°C.
   b. 0°C.
   c. −20°C.
   d. −40°C.
   e. none of the above.
7. High-intensity focused ultrasound (HIFU) exerts what effect on prostate tissue?
   a. Tissue fragmentation with disruption of vascular architecture
   b. Coagulative necrosis
   c. Nuclear injury
   d. Cavitation
   e. Disruption of protein synthesis
8. There exists level 1 evidence that prostate focal therapy:
   a. is equivalent to radical therapy in terms of cancer control.
   b. is superior to active surveillance for progression free survival.

c. is superior to active surveillance for overall survival.
d. is equivalent to active surveillance for overall survival.
e. is superior to radical therapy in terms of sexual function outcomes.

9. The following modalities all employ a thermal form of ablation EXCEPT:
    a. MRI-guided transurethral ultrasound ablation.
    b. focal laser ablation.
    c. focal cryotherapy.
    d. vascular-targeted photodynamic therapy.
    e. none of the above.
10. Reliable markers of post-focal therapy treatment success include:
    a. post-treatment PSA levels.
    b. digital rectal examination.
    c. post-treatment mpMRI alone.
    d. targeted biopsy of the treated area only.
    e. targeted biopsy of the treated area plus a systematic biopsy.

Answers

1. **c. It accurately estimates lesion size for determination of ablation margin.** mpMRI has been consistently shown to underestimate the actual tumor volume by 5% to 30% in historadiological correlation studies.
2. **e. All of the above.** The oncological premise of focal therapy is to completely destroy the index lesion while preserving as much function as possible. As such, selection of the ablative technology will depend on availability, operator experience, desired approach, and the suitability of the technology to deliver a lethal dose of energy to the desired location in order to successfully ablate the lesion.
3. **c. Warming of neurovascular bundles.** Use of transrectal ultrasonography (TRUS) for real-time monitoring of the freezing process, use of a urethral warming catheter, use of thermocouples placed in critical areas of the prostate, and improved cryoprobes

allowing percutaneous insertion are technical innovations that have all contributed to improving cryotherapy for prostate cancer.

4. **b. The number of freeze/thaw cycles.** In a clinical setting, the number of freezing cycles, the lowest temperature achieved, and the existence of any regional "heat sinks" may be more important factors relating to cancer destruction. Repeating a freeze/thaw cycle results in more extensive tissue damage compared with a single cycle.
5. **d. Hypoechogenicity with a hyperechogenic rim.** Frozen tissue is significantly different from unfrozen tissue in sound impedance, resulting in strong echo reflection at the interface of frozen and normal tissue. The frozen area can be seen as a well-marginated hyperechoic rim with acoustic shadowing by ultrasonography. Sonography provides no information about the temperature distribution within the ice, nor does it show the extent of freezing at the lateral or anterior aspects of the prostate.
6. **d. − 40°C.** Complete cell death is unlikely to occur at temperatures higher than −20°C, and temperatures lower than −40° C are required to completely destroy cells.
7. **b. Coagulative necrosis.** Highly focused sound energy results in mechanical and thermal effects on tissue. In the case of HIFU, the primary mechanism of tissue ablation is by raising the temperature in the tissue above the level needed to create coagulative necrosis.
8. **b. Is superior to active surveillance for progression free survival.** Azzouzi et al. randomized 415 men with low-risk prostate cancer to focal VTP or active surveillance demonstrating that FT significantly improved negative biopsy at 2 years and decreased disease progression and those requiring definitive disease intervention. Of the 216 men assigned to VTP, 196 received the treatment and 195 completed 24 months of follow-up. In the AS arm, 174 men completed 24 months of follow-up. Overall, cancer progression rate (volume, grade, or PSA progression) in the AS arm was 58% compared to 28% in the VTP arm. This is the first randomized trial with level 1 evidence comparing focal therapy to an accepted management strategy demonstrating a meaningful clinical benefit.

9. **d. Vascular-targeted photodynamic therapy.** In VTP, low-power near-infrared laser light of specific wavelength destroys targeted tissues that have been photosensitized with an intravascular agent preferentially taken up by tumor cells. This occurs via the production of reactive oxygen species in the target zone causing irreversible damage to cell membranes and small arterioles leading to tumor necrosis.
10. **e. Targeted biopsy of the treated area plus systematic biopsy.** A 2015 ICUD consensus on follow-up after focal therapy recommended an mpMRI with mandatory targeted biopsy of 4 to 6 cores in the treated area at 3 to 6 months, then mpMRI at 1 to 2 and 5 years with targeted biopsy as needed, especially if a new lesion should become manifest. A systematic biopsy was also recommended at 1 to 2 and 5 years to provide further histological evaluation of the untreated zone for potential outfield recurrence.

Chapter Review

1. Gleason patterns 4 and 5 demonstrate many of the hallmarks of malignancy compared to Gleason pattern 3 and represent priority targets for elimination. The feasibility of focal therapy for a multifocal disease such as prostate cancer is based on the presumption that an index lesion is primarily responsible for disease progression and metastases.
2. The index lesion as the focus of disease progression and/or metastases is supported by: (1) most of the time, the grade/stage of the cancer is determined by the index lesion, (2) most of the tumor volume is often contributed by the index lesion, (3) majority of satellite tumors are small (<0.5 mL) and low grade, and (4) genetic studies suggest a monoclonal origin of metastatic or lethal prostate cancer.
3. A successful focal therapy strategy depends on a multidisciplinary team to accurately determine the disease extent/location through advanced imaging and biopsy, ascertain that the patient will be compliant to follow-up, completely ablate the index lesion(s), and

monitor the patient post-treatment with a view to future targeted treatment of either persistent or de novo disease, or conversion to whole-gland treatment as necessary.

4. The use of TRUS-guided biopsy to guide focal therapy is fraught with the following difficulties: it overdiagnoses clinically insignificant cancer; it misses clinically significant cancers in 30% of cases—usually in the anterior or apical regions; it may underrepresent true cancer burden; and repeat biopsies do not give consistent results.
5. mpMRI has become a key enabler for prostate focal therapy due to its ability to preferentially detect high-grade lesions, improve the detection of anterior zone cancers, improve the detection of extra-prostatic extension, and accurately identify the index lesions for targeted biopsy. However, systematic biopsy remains necessary to evaluate the prostate gland prior to focal therapy due to the 10% to 15% miss rate with mpMRI. Other imaging modalities such as multiparametric ultrasound or PSMA-PET CT/MRI have potential use in this area but remain to be further evaluated.
6. Focal therapy can be applied in various patterns (hemi-ablation, quadrant-ablation, lesion-ablation) using different technologies (cryotherapy, HIFU, IRE, laser, etc.). The chosen approach depends on the location of the tumor, the desired ablative technology that is available, and other anatomical considerations that may be peculiar to the patient.
7. Cryotherapy is a thermal ablative modality achieving cell kill through extraction of heat producing lethal cold temperatures. Cryosurgery kills target tissues via physical and physiological changes including mechanical forces resulting in membrane disruption created by intra- and extracellular ice formation, recrystallization, vascular stasis/ischemia, apoptosis, and immune effects.
8. Recent developments have made cryotherapy increasingly suited for focal therapy. First, new third-generation cryoprobes are as fine as 17-gauge, allowing them to be maneuvered to the appropriate position within the prostate. Second, variable ice length adjustment allows the ice ball to be contoured exactly to the lesion that is to be ablated. Third, the ice edge can be clearly seen using transrectal

ultrasound allowing for the extent of ablation and the ablation margin to be monitored and adjusted in real time. Fourth, the use of a urethral warmer has greatly reduced the risk of urethral complications arising from cold injury to the urethral mucosa.

9. Cell destruction is determined by the cooling rate, warming rate, and the nadir temperature attained. Slow thawing is generally more effective in tissue destruction compared to rapid thawing. A nadir temperature of −40°C and a double freeze/thaw cycle are recommended for adequate cell death in the target zone.
10. High-intensity focused ultrasound (HIFU) is a process where multiple ultrasound beams are focused onto a preset point generating a temperature of at least 55°C to produce coagulative necrosis within the desired target. The transducer is typically built into the transrectal ultrasound probe which is also used for treatment planning and monitoring making it well suited for treatment of posterior zone lesions. However, the anterior gland may be more difficult to treat due to energy dissipation over the intervening prostate tissue and displacement of anterior zone targets with gradual edema of the prostate tissue as treatment progresses.
11. In-bore MRI guided HIFU (MRgFUS) and MRI-guided transurethral ultrasound ablation of the prostate (TULSA) are new developments being actively studied.
12. Focal laser ablation (FLA) refers to the creation of coagulative necrosis using an interstitially placed laser fiber. The ablation zone is guided by the laser fiber, and tissue destruction occurs only within a fixed distance from the fiber tip, offering the possibility of true focal ablation where accurate and predictable tissue destruction zones can theoretically be safe within close proximity to vital structures. The disadvantages are that longer procedure times or extra laser fibers may be necessary for ablation of larger lesions, and early studies show that additional care may be needed to treat the lesion with an adequate margin.
13. Irreversible electroporation (IRE) is a non-thermal ablative technique that uses short pulses of direct current electricity to produce irreversible pores in the cell membrane, leading to cell death, and preferentially damages cells while preserving connective tissue

architecture that could theoretically serve as a scaffolding for new cells to grow.

14. Vascular-targeted photodynamic therapy (VTP) employs the use of a low-power near-infrared laser light of specific wavelength producing reactive oxygen species in tissues that have been photosensitized with an intravascular agent preferentially taken up by tumor cells causing irreversible damage to cell membranes and small arterioles leading to tumor necrosis.
15. A randomized trial (Azzouzi et al.) of 415 men with low-risk prostate cancer compared focal VTP or active surveillance demonstrating that FT significantly improved negative biopsy at 2 years and decreased disease progression and those requiring definitive disease intervention. Of the 216 men assigned to VTP, 196 received the treatment and 195 completed 24 months of follow-up. In the AS arm, 174 men completed 24 months of follow-up. Overall, cancer progression rate (volume, grade, or PSA progression) in the AS arm was 58% compared to 28% in the VTP arm. This is the first randomized trial with level 1 evidence comparing focal therapy to an accepted management strategy demonstrating a meaningful clinical benefit.
16. Brachytherapy, stereotactic body radiotherapy (SBRT), toxins, and various nanoparticles are currently being investigated for use in prostate focal therapy.
17. After prostate focal therapy, PSA, which is a traditional marker of therapeutic success, is less relevant due to its levels being affected by the amount of residual prostate epithelia that may continue to grow. MRI and prostate biopsies are recommended in order to assess the treated (infield) area as well as the untreated (outfield) area.
18. Future research endeavors should focus on better patient selection using clinical, imaging, and/or genetic biomarkers, cost-effectiveness studies, and the investigation of adjuvant agents to improve the efficacy of focal therapy in prostate cancer.